Getting started with CompetingRiskRPackage

1. Data and a multivariable fit

CompetingRiskRPackage fits weighted Fine-Gray proportional subdistribution hazards models for case-cohort competing-risk data. A single fit provides coefficient estimates and three proportionality diagnostics: **cor**, **rank cor** and **reg**.

The example uses a supplied synthetic dataset of 4,000 subjects. A two-level factor is added to demonstrate fitting continuous and categorical covariates together.

```
library(CompetingRiskRPackage)
dat <- read.csv(system.file("extdata", "example_ph_4000.csv",
                            package = "CompetingRiskRPackage"))
set.seed(2026)
dat$group <- factor(sample(c("A", "B"), nrow(dat), replace = TRUE))
dat$id <- seq_len(nrow(dat))
head(dat, 3)
#>      time status subcohort covariate1 group id
#> 1 0.42621      1         0     0.6987     A  1
#> 2 0.02541      1         1     0.5565     A  2
#> 3 0.08179      1         1     0.1401     A  3
```

The input data must contain one row per full-cohort subject, including subjects with unmeasured covariates. Specify the following columns:

- `time`: finite, non-negative observed follow-up time.
- `status`: 0 for censoring; positive integers identify event causes. Here, 1 is the target event and 2 is the competing event.
- `subcohort`: 1 (or `TRUE`) for random subcohort membership, 0 otherwise. Subcohort membership is specified independently of covariate measurement availability.

Specify all covariates in a single one-sided formula. Categorical covariates use R's model-matrix contrasts, and the model intercept is omitted. Here, `groupB` represents level B relative to reference level A.

```
fit <- cc_fit(~ . - id, data = dat,
              time = "time", status = "status", subcohort = "subcohort",
              cause = 1)
c(cohort = fit$n, target_events = fit$n_events,
  subcohort = fit$n_subcohort)
#>        cohort target_events     subcohort
#>          4000          2713          2428
summary(fit)$coefficients
#>             estimate SE_sandwich SE_information
#> covariate1  0.62755     0.08044        0.06812
#> groupB      0.02709     0.04539        0.03841
```

`summary(fit)$coefficients` reports adjusted coefficient estimates, sandwich standard errors (`SE_sandwich`) and information-based standard errors (`SE_information`).

The current implementation supports unstratified random subcohort sampling with time-fixed covariates; it does not support stratified sampling, delayed entry or time-varying covariates.

2. Results and fitting all covariates

```
fit$tests[, c("method", "covariate", "statistic", "df", "p.value", "status")]
#>     method  covariate statistic   df p.value status
#> 1      cor covariate1   -0.3110 2711  0.7558     ok
#> 2      cor     groupB    1.4679 2711  0.1422     ok
#> 3 rank cor covariate1   -0.1414 2711  0.8876     ok
#> 4 rank cor     groupB    1.4585 2711  0.1448     ok
#> 5      reg     GLOBAL    2.0402    2  0.3606     ok
#> 6      reg covariate1    0.0903    1  0.7638     ok
#> 7      reg     groupB    2.0069    1  0.1566     ok
```

The **cor** and **rank cor** diagnostics correlate each residual column with event time and ranked event time, respectively. Rank cor uses average ranks for tied times and leaves residuals unranked. Their p values use Student t reference distributions. **reg** is a nuisance-adjusted score test using a chi-squared reference distribution; the default time function is `function(t) t`.

The `GLOBAL` reg row reports the joint test across all coefficients, with 2 degrees of freedom in this example. Additional rows report coefficient-specific tests. Categorical covariates are reported by their expanded coefficients; a separate joint test for each categorical covariate is not provided.

The reported p values test the proportionality assumption for the corresponding covariate or coefficient set. `summary(fit)` combines coefficient estimates, diagnostics and fit status.

Including all covariates. Use `~ .` to include all columns except those specified by `time`, `status` and `subcohort`. The formula `~ . - id` additionally excludes the subject identifier. Additional columns can be excluded using formula subtraction, for example `~ . - id - visit`. An explicit covariate formula gives the same fit:

```
fit_named <- cc_fit(~ covariate1 + group, dat, "time", "status", "subcohort")
isTRUE(all.equal(coef(fit), coef(fit_named)))
#> [1] TRUE
```

Selecting residual plots. `plot(fit)` displays residual plots for all coefficients. Use `covariates` to select variables or expanded coefficients:

```
plots <- plot(fit, covariates = c("covariate1", "group"))
names(plots)
#> [1] "covariate1" "groupB"
```

Plot selection preserves the fitted model and its covariate adjustment. A factor name selects all its non-reference coefficients; an expanded name, such as `"groupB"`, selects just that coefficient. The method invisibly returns a named list of plot objects. Display an individual plot with `plots[["covariate1"]]`.

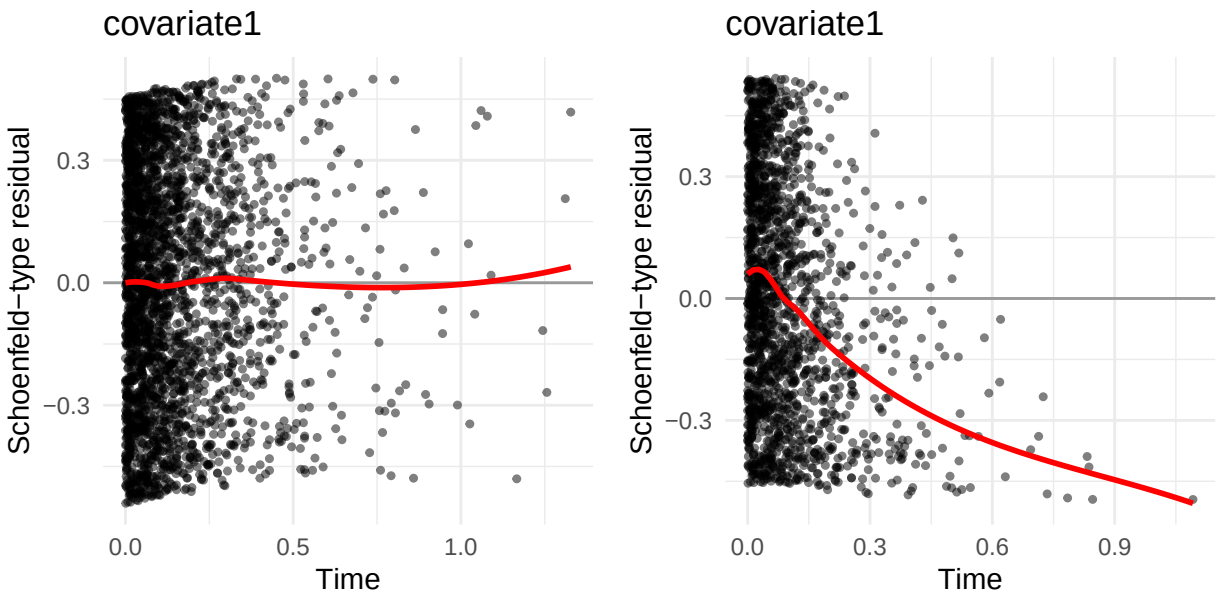
3. Residual plots from multivariable models

The following example compares two synthetic datasets of 4,000 subjects. The first has constant covariate effects; the second has a time-varying effect for `covariate1`. Both models include `covariate1` and `group`.

```
nonph <- read.csv(system.file("extdata", "example_nonph_4000.csv",
                              package = "CompetingRiskRPackage"))
nonph$group <- dat$group
nonph$id <- dat$id
fit_nonph <- cc_fit(~ . - id, nonph,
                    "time", "status", "subcohort")
```

Select `covariate1` to display its residual plot from each multivariable fit, adjusted for `group`. The two plots are arranged side by side below:

```
plot(fit, covariates = "covariate1")
plot(fit_nonph, covariates = "covariate1")
```



PH-generated data (left) and non-PH-generated data (right). Each panel displays the `covariate1` residuals against target-event time, with a horizontal reference at zero and a LOESS smooth. Axis limits are determined separately for each plot.

Use `plot(fit, covariates = "group")` to display the factor's residual plots, or `plot(fit)` to display all coefficient plots.

4. Additional options and result extraction

Time transformation. Set `time_transform` to specify the time function used by reg. The example below uses a quadratic function; cor and rank cor are unchanged.

```
fit_quad <- cc_fit(~ . - id, dat, "time", "status", "subcohort",
                   time_transform = function(t) t^2)
fit_quad$tests[fit_quad$tests$method == "reg",
               c("covariate", "statistic", "df", "p.value", "status")]
#>    covariate statistic df p.value status
#> 5     GLOBAL    1.2555  2  0.5338     ok
#> 6 covariate1    0.0897  1  0.7646     ok
#> 7     groupB    1.2548  1  0.2626     ok
```

Missing covariates. Covariates are required for subcohort members and target-event cases. Missing values are permitted for other subjects, whose follow-up and status must remain in the dataset. The example below introduces permitted missing values and reproduces the coefficient estimates:

```
dat_missing <- dat
unrequired <- dat$status != 1 & dat$subcohort == 0
dat_missing[unrequired, c("covariate1", "group")] <- NA
fit_missing <- cc_fit(~ . - id, dat_missing, "time", "status", "subcohort")
isTRUE(all.equal(coef(fit), coef(fit_missing)))
#> [1] TRUE
```

Result extraction. `coef()` returns coefficient estimates, `vcov()` returns the complete sandwich covariance matrix, and `residuals()` returns a matrix with one column per coefficient:

```
b <- coef(fit)
V <- vcov(fit)
r_full <- residuals(fit)                 # All rows; NA on non-target rows
r_event <- residuals(fit, type = "event") # Target events in input order
dim(V)
#> [1] 2 2
dim(r_full)
#> [1] 4000    2
c(converged = fit$converged, covariance_ok = fit$variance_status == "ok")
#>     converged covariance_ok
#>          TRUE          TRUE
```

Full residuals preserve input row order, with `NA` for non-target-event rows. Event residuals correspond to `fit$event_index` and `fit$event_times`; tied events retain separate rows.

Fit status. Check `fit$converged`, `fit$variance_status`, and the `status` and `message` columns in `fit$tests`. Unavailable inference is returned as `NA`; coefficient estimates and descriptive residuals may remain available after a covariance failure. A reg-specific failure does not necessarily affect cor or rank cor.

Further reading and versions. See `?cc_fit` and `?cc_fit_methods` for the interface. Methodological background: Wogu et al., *Assessing the Proportional Hazards Assumption of Fine-Gray Model in Case-Cohort Studies* (manuscript). Rendered with CompetingRiskRPackage 0.3.7 and R 4.3.0.